

Lab05

AUTHOR

Pedro José

Criando a função

```
analisa_aeronave <- function(tail_number, arquivo) {

  # Lendo a tabela de aeroportos
  airports <- read_csv("airports.csv")

  # 1 - Lendo o arquivo em chunks e filtrando o avião
  voos_do_aviao <- read_csv_chunked(
    arquivo,
    callback = DataFrameCallback$new(function(chunk, pos) {
      chunk %>% filter(TAIL_NUMBER == tail_number)
    }),
    chunk_size = 1000000
  )

  # 2 - Ordenando por data e hora
  voos_do_aviao <- voos_do_aviao %>%
    arrange(YEAR, MONTH, DAY, SCHEDULED_DEPARTURE)

  # 3 - Juntando com Airports

  voos_do_aviao <- voos_do_aviao %>%
    left_join(airports, by = c("ORIGIN_AIRPORT" = "IATA_CODE")) %>%
    rename(
      LATITUDE_ORIGEM = LATITUDE,
      LONGITUDE_ORIGEM = LONGITUDE
    )

  # 4 - Juntando com Airports

  voos_do_aviao <- voos_do_aviao %>%
    left_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE")) %>%
    rename(
      LATITUDE_DESTINO = LATITUDE,
      LONGITUDE_DESTINO = LONGITUDE
    )

  # 5 - criando o mapa

  mapa <- leaflet(voos_do_aviao) %>%
    addTiles() %>%
    addPolylines(
      lng = ~c(LONGITUDE_ORIGEM, LONGITUDE_DESTINO),
      lat = ~c(LATITUDE_ORIGEM, LATITUDE_DESTINO),
      group = "Trajeto"
    ) %>%
    addMarkers(
      lng = ~LONGITUDE_ORIGEM,
```

```

lat = ~LATITUDE_ORIGEM,
popup = ~paste(
  "Aeroporto:", ORIGIN_AIRPORT, "<br>",
  "Data:", DAY, "/", MONTH, "/", YEAR
),
group = "Aeroportos"
)

# 6 - Retornando a lista

return(list(tabela = voos_do_aviao, grafico = mapa))
}

```

Testando a função

```
library(tidyverse)
```

Warning: pacote 'tidyverse' foi compilado no R versão 4.5.3

```

— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.0    ✓ stringr    1.5.1
✓ ggplot2    4.0.3    ✓ tibble     3.2.1
✓ lubridate  1.9.4    ✓ tidyr      1.3.1
✓ purrr      1.0.4

— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

```

```
library(leaflet)
```

Warning: pacote 'leaflet' foi compilado no R versão 4.5.3

```

library(readr)

resultado <- analisa_aeronave("N431WN", "flights.csv.zip")

```

Rows: 322 Columns: 7

```

— Column specification —
Delimiter: ","
chr (5): IATA_CODE, AIRPORT, CITY, STATE, COUNTRY
dbl (2): LATITUDE, LONGITUDE

```

ℹ Use `spec()` to retrieve the full column specification for this data.
 ℹ Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```

— Column specification —
cols(
  .default = col_double(),
  AIRLINE = col_character(),

```

```

TAIL_NUMBER = col_character(),
ORIGIN_AIRPORT = col_character(),
DESTINATION_AIRPORT = col_character(),
SCHEDULED_DEPARTURE = col_character(),
DEPARTURE_TIME = col_character(),
WHEELS_OFF = col_character(),
WHEELS_ON = col_character(),
SCHEDULED_ARRIVAL = col_character(),
ARRIVAL_TIME = col_character(),
CANCELLATION_REASON = col_character()
)

```

❗ Use `spec()` for the full column specifications.

Warning in validateCoords(lng, lat, funcName): Data contains 156 rows with either missing or invalid lat/lon values and will be ignored

Visualizando a função

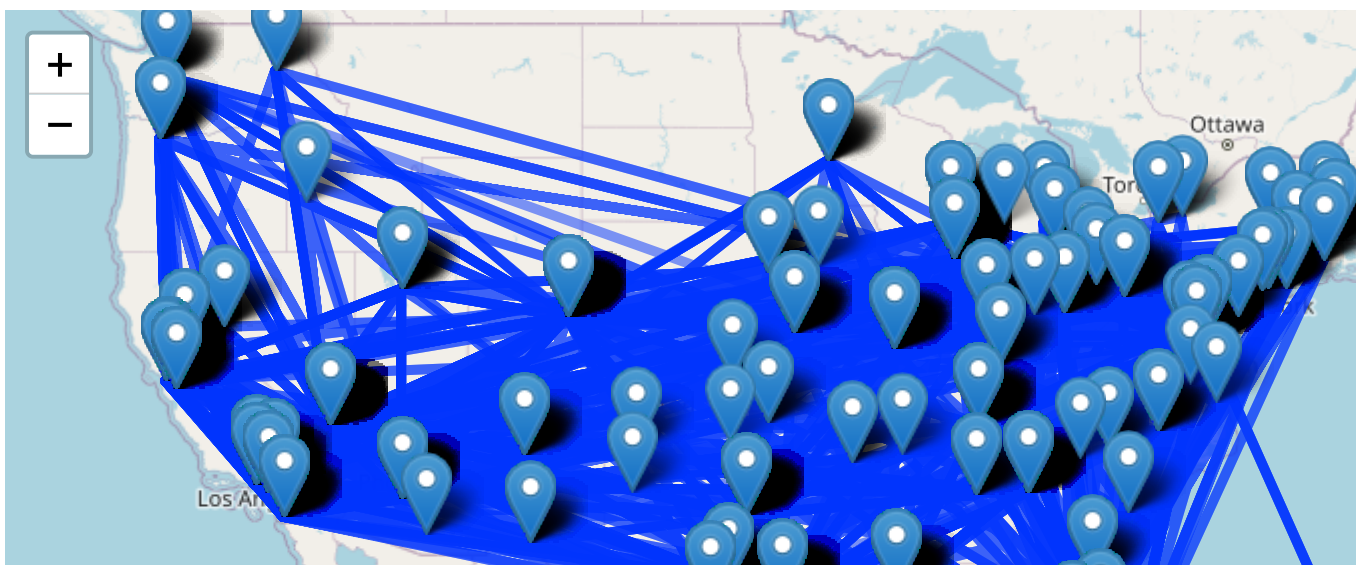
```
head(resultado$tabela)
```

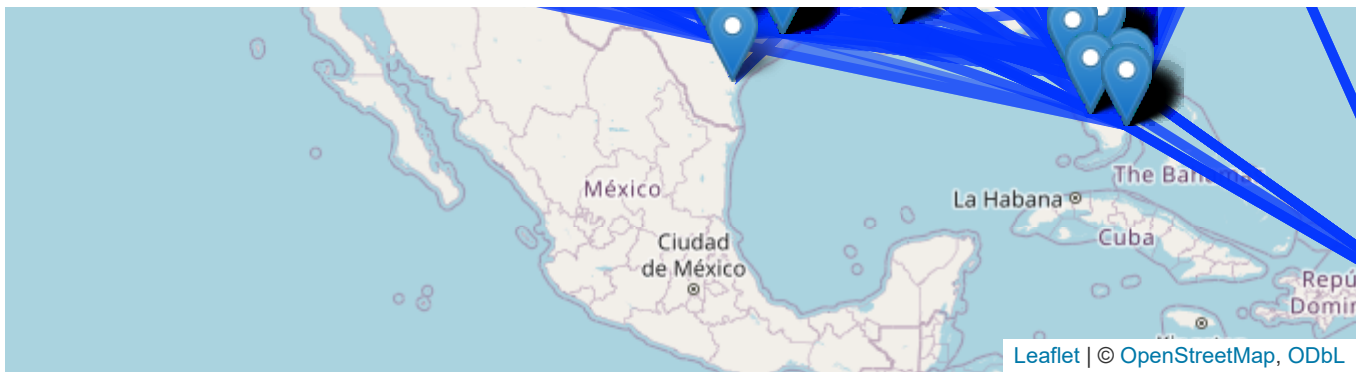
A tibble: 6 × 43

	YEAR	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT
	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<chr>	<chr>
1	2015	1	1	4	WN	1762	N431WN	DCA
2	2015	1	1	4	WN	3564	N431WN	HOU
3	2015	1	1	4	WN	3564	N431WN	BOS
4	2015	1	1	4	WN	3564	N431WN	BWI
5	2015	1	2	5	WN	1793	N431WN	MCI
6	2015	1	2	5	WN	1793	N431WN	LAX

i 35 more variables: DESTINATION_AIRPORT <chr>, SCHEDULED_DEPARTURE <chr>,
DEPARTURE_TIME <chr>, DEPARTURE_DELAY <dbl>, TAXI_OUT <dbl>,
WHEELS_OFF <chr>, SCHEDULED_TIME <dbl>, ELAPSED_TIME <dbl>, AIR_TIME <dbl>,
DISTANCE <dbl>, WHEELS_ON <chr>, TAXI_IN <dbl>, SCHEDULED_ARRIVAL <chr>,
ARRIVAL_TIME <chr>, ARRIVAL_DELAY <dbl>, DIVERTED <dbl>, CANCELLED <dbl>,
CANCELLATION_REASON <chr>, AIR_SYSTEM_DELAY <dbl>, SECURITY_DELAY <dbl>,
AIRLINE_DELAY <dbl>, LATE_AIRCRAFT_DELAY <dbl>, WEATHER_DELAY <dbl>, ...

```
resultado$grafico
```





Chunk para ver o horário

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-16 16:24:32 -03"
```